Bischoff 2021 LUAD — analysis summary

GO / NO-GO GATES

Gate 1 — integration & continuity → GO

Malignant structure is mixed and continuous on one manifold. Judged on the embedding + mixing metrics across an n_pcs×θ sweep: θ=0 gives per-sample islands (iLISI 1.05, diff-sample kNN 0.02) — integration mixes them strongly, saturating by θ=2 (iLISI ~3.3–3.7 across θ=2/5/10). At the chosen θ=10 (Kürten precedent) the kNN graph is a single connected component (comp=1, as at every θ) and strongly mixed (iLISI 3.53, diff-sample kNN 0.43 of 0.805 ceiling). The two density lobes are poles of one connected manifold — not disconnected islands (comp=1 throughout) and not sample-driven (both multi-sample); the thin transitional waist is expected differentiation density, not a break. Cell cycle does not structure the manifold (≤6% cycling per lobe) — no compartment to strip.

Gate 2 — embryonic module pattern → GO (non-branching, flagged)

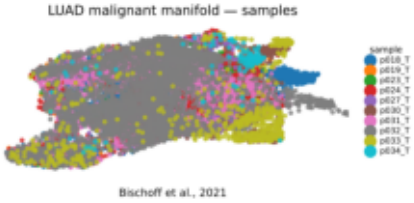
Non-branching case (§1): Crest is a clean inverse/terminal gradient — anti-correlates Gast (−0.42) and pSom (−0.50), and the Gast&pSom co-high hub is strongly Crest-low (z −0.82 vs +0.20). Flag (pass anyway): pSom is near-ubiquitous (score>0 in 94% of cells), Gast sparse (14%), Gast/pSom co-variation weak (−0.15) — an attenuated signal versus the Kürten precedent. The axis is alveolar-differentiation ↔ EMT.

Gate 3 — bifurcation cross-repression → NA

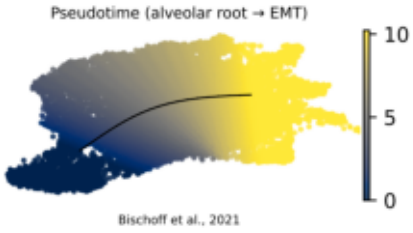
No bifurcation. The curve is fold-free (2 tips, 0 forks, self-approach 2.9); the PPT tree's forks separate no biology (all tips Gast≈−0.035 / pSom≈+0.06 / Crest≈−0.035). Branching-only gate → NA.

STEP FIGURES

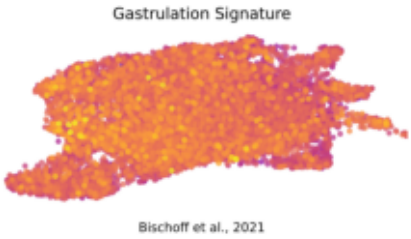
Manifold — samples



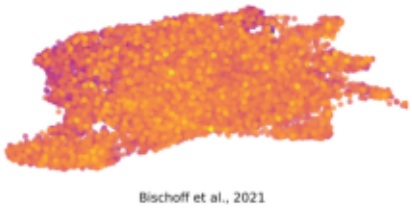
Pseudotime + curve



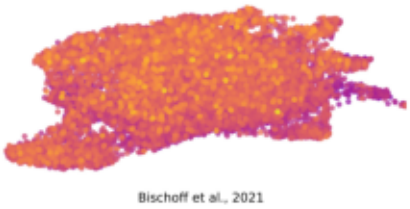
Gastrulation



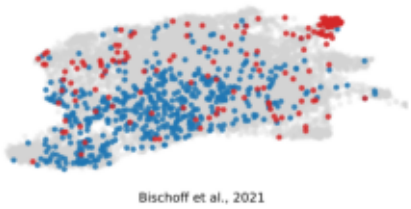
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Lung adenocarcinoma (Bischoff et al. 2021), 10x scRNA-seq. 13,382 malignant cells across 10 tumor samples (of 20; normal-tissue samples carry 2–15 malignant cells and drop out at threshold).

Verdict.
The embryonic pattern reproduces in the non-branching form — a single continuous alveolar→EMT manifold with Crest as an inverse terminal gradient and no Gast/pSom bifurcation. The signal is weaker than the Kürten HNSCC precedent (attenuated; flagged), but present and continuous.

Parameter choices

Min-cell threshold	43 (geo-mean across the 127→15 gap, ratio 8.5) → 10 samples, 13,382 cells	**Islands removed**	0 (no detached components at n_neighbors=30) → 13,382 on the manifold
Overdispersed genes	1,976 (no pre-PCA scaling)	**Integration / embedding**	Harmony on sample θ=10; n_pcs=20, n_neighbors=30, UMAP spread=1.3 (seed 1)
Geometry	2-D continuum (UMAP aspect ≈1.3); scFates curve (non-branching)	**Root**	Alveolar / Crest-low tip (Gast/pSom-high pole; Crest is developmentally late)
EMT-TFs shown	FOXC1, TCF12, TCF3, SNAI1, PRRX1/2, ZEB1/2, TCF4 (≥3% detected; VIM/TGFBI not TFs)	**Gast–pSom r (whole manifold)**	−0.15 raw — attenuated vs Puram hub ref ≈ −0.49 (no windowed hub step; Gate 3 NA)